Memo

To: Sarah Okafor, Head of E-Commerce | **From:** Data & ML Team | **Date:** 13 October 2026
Subject: "Customers who bought this also bought" suggestions, and our main customer groups

Bottom line. We can shrink the data from about 180 MB to 7.6 MB and run the whole nightly job in about 1.1 seconds. On our test, personalised suggestions **beat best sellers**: they matched 39% of customers, best sellers 27%. We recommend a live trial. For marketing we found three customer groups by what they buy, each split into UK and international customers.

1. Product-page suggestions

- **Method and why.** We used a standard technique called Truncated SVD. It boils each customer's and each product's buying history down to 100 numbers, so products bought by the same kinds of customers end up with similar numbers. We chose it because our data is almost all blank (98% of customer-product pairs) and this method works on it directly. The best-known alternative (PCA) gave the same results within the margin of error.
- **Size and speed.** The full customer-by-product table is about 180 MB. The compressed version is 7.6 MB (23 times smaller); the product-page lists alone need 3.3 MB. The whole nightly job takes 1.1 seconds: 0.7 to build the model, 0.2 for the "also bought" list of every product, 0.3 to pick for every customer. We did not time the old prototype, so we cannot quote a speed-up.
- **What we lose.** The 100 numbers keep the broad buying patterns (about 47% of the detail in the data) but cannot rebuild any one customer's exact shopping list. 200 numbers keep more (59%) but suggested no better.
- **Are they better than best sellers? Yes, on our test.** We hid the last two months of purchases (9 Oct to 9 Dec 2011) and checked whether any of a customer's 10 suggestions was a product they then bought for the first time. Suggestions hit for **38.9%** of customers; the 10 best sellers hit for **26.8%**, so 45% more customers. The gap is far bigger than chance, and an earlier two-month period gave the same picture (40.7% vs 25.9%). Our first version, which ignored how popular a product is, lost (about 21%); keeping popularity in the score fixed that. Settings were chosen on the earlier period, not on the test.
- **One caution.** The test scored picks based on a customer's own history. The "also bought" list on a product page uses the same model but was not scored separately, so it needs its own live test.

2. Customer groups for marketing

Group	Customers	Products bought (UK / outside)	What stands out (UK)	Outside the UK	Campaign idea (untested)
Bags & baking	1,300 (24%)	72 / 63	Jumbo and lunch bags (jumbo bag: 50% buy it vs 16% overall)	Germany 23%, France 21%; cake cases and charlotte bags	Bags for UK shops; cake cases and charlotte bags, in German and French, abroad
Hearts & home décor	1,534 (28%)	45 / 46	Heart T-light holders (57% vs 26%), wicker hearts, zinc heart decorations	Only 44 customers: too few for their own campaign	New hearts and candle lines for UK shops
Light buyers	2,637 (48%)	30 / 37	Vintage toys and games (a mild theme)	Kids' accessories (snack boxes, plasters in tins); Germany 20%, France 17%	Kids' accessories for international light buyers

Groups are formed from what customers buy, with shop size removed; the UK / outside-UK split uses each customer's main country. The picture is a simplification, and the light buyers have the weakest theme. No customer IDs appear in this memo or in any figure.

3. What could go wrong

- **Our test may be too tough or too narrow.** It covers the pre-Christmas months and counts only products a customer had never bought before, so it says nothing about reorders.
- **Gaps in what the model can see.** New customers and new products cannot be suggested until the nightly job has seen them (about 15% of customer-product pairs in the test period came from customers not in our training data). Products with fewer than 5 buyers (10% of the catalogue, 0.2% of sales lines) are left out, and the 22% of sales lines with no customer ID are not in the model.

4. Next steps

- Run a live A/B test of personalised suggestions and of the product-page list against best sellers.
- Test one campaign per group, and fold the 44-customer group into a larger campaign.
- Compare the "ever bought" version of the data with the current one on the earlier period; it looked as good on the test.